LESSON PLAN

NDT (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Optical holography basic principle, procedure for setting up construction and re-construction, laser, interferogram advantages disadvantages
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Optical holography basic principle, procedure for setting up construction and re-construction, laser, interferogram advantages disadvantages and application
5. Specific Objectives	19.1.5.1 Trainee will be able to explain the optical holography and acoustic emission inspection

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of holographic setup using laser source, beam splitter, and recording medium; safety precautions regarding laser eye protection and alignment stability; instructor supervision of beam path and recording process.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the fundamental difference between a photograph and a hologram?
(b) Name one type of laser commonly used in non-destructive testing.
(c) Why is coherence essential for interferometry?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principle of optical holography • Role of laser coherence and monochromaticity • Interference pattern formation • Recording medium requirements	L & Ds	PPT, IB	15 min
(b)	• Procedure for setting up construction • Laser alignment and beam splitting • Reference and object beam configuration • Interferogram advantages and disadvantages	L & Ds	PPT, PDF	20 min
(c)	• Procedure for re-construction of hologram • Light source requirements for reconstruction • Image formation and viewing angles • Applications in NDT	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the interference pattern stores three-dimensional information.
(b) List two advantages and two disadvantages of using interferograms in inspection.
(c) Describe the critical steps required to successfully reconstruct a holographic image.

12. Home Assignment and Follow Up	Prepare a brief report comparing the setup requirements for optical holography and acoustic emission inspection, highlighting key differences in data acquisition.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Acoustic emission, basic concept, types, application, advantages	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min